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Epibiotic association of encrusting cheilostome bryozoans on shells of an invasive mussel from rocky shores of South Africa, with the description of a new aviculiferous species of *Chaperia*

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Abstract

Rocky shores typically exhibit a variety of sedentary and free-moving forms of epibionts associated with the shells of mussel basibionts. This paper provides a first report on epibiotic bryozoans found on shells of the invasive Mediterranean mussel, *Mytilus galloprovincialis* Lamarck. More than 2500 mussels were collected between December 2019 and October 2020 from rocky shores during low spring tides across the south–southeast coast of South Africa. Ten percent of these mussels hosted epibiotic bryozoans. We examined a subset of these epibiotised mussels to assess the diversity of bryozoans. Three encrusting cheilostome species were identified: *Chaperia atypica* n. sp., *Celleporella hyalina* (Linnaeus), and *Hippomonavella* sp. This new species is the first *Chaperia* with avicularia and the first South African species with ooecia. This study highlights the biological diversity of epibiotic bryozoans on mussel shells and, given their differences in microtopography, the possibility that invasive species can provide a new substratum for rare, overlooked or undescribed species of epibionts.

Key words: *Celleporella*, epibiosis, *Hippomonavella*, intertidal mussels, *Mytilus galloprovincialis*, new species

Introduction

Epibionts (i.e. organisms living on another living organism) are common in marine environments, span numerous taxonomic divisions and phyla, occur on a wide taxonomic diversity of basibionts (hosts), and can either be host-specific (obligate) or host non-specific (facultative; Thornber *et al.* 2016). Bryozoans also referred to as ‘moss animals’ or ‘sea mats’, are commonly reported as epibionts and typically attach to various substrata such as rocks, shells, algae, and other bryozoans (e.g. Bock 1982; Key *et al.* 1996; Hayward & Ryland 1999; Key & Hendrickx 2022). Epibiosis can have positive effects for both the host basibiont and the epibiont, such as increased fecundity (e.g. Barea-Arco *et al.* 2001), or negative effects, such as decreased fecundity (e.g. Kumari *et al.* 2021; Pan *et al.* 2021) or interference with feeding and locomotion (e.g. Ramos-Rivera *et al.* 2021; Kumar *et al.* 2022).

In South Africa, all prior taxonomic works focus mainly on the diversity of bryozoans (Boonzaaier-Davids *et al.* 2020), but few have investigated the epibiont-basibiont relationship including diversity and abundance. An interesting example of bryozoan epibiosis found in South Africa is that between the ctenostome bryozoan *Alcyonidium nodosum* O’Donoghue & de Watteville, 1944 and the subtidal whelk *Burnupena papyracea* (Bruguère, 1789). This shell-encrusting bryozoan species produces a toxic chemical that protects *B. papyracea* against predators, like the rock lobster *Jasus lalandii* (H. Milne Edwards, 1837) (see Barkai & McQuaid 1988; Ryland 2001; Gray *et al.* 2005; Florence *et al.* 2007).

On rocky shores, a variety of sedentary and free-moving forms of epibionts are associated with the shells of mussel basibionts. For instance, epibionts on shells of the invasive mussel, *Mytilus galloprovincialis* Lamarck,

1819, in South Africa include algae, sponges, sea anemones, barnacles, spirorbid worms, limpets, chitons, mussel spat, and encrusting bryozoans (Thieltges 2005; Bell *et al.* 2015; Ma *et al.* 2021). Introduced to South Africa in the late 1970s, *M. galloprovincialis* has since spread to rocky shores across the region from the southeast coast of South Africa (i.e. the southern coast of the Transkei) to northern Namibia (Ma *et al.* 2021). Individual epibionts can be found on specific areas of the host and their diversity and abundance can depend on the ontogenic stage of the basibiont (Wahl 2009). The microtopography of surfaces has particular implications for fouling (Bers & Wahl 2004) as the periostracum, i.e. the outermost, proteinaceous layer of mussel shells, has anti-fouling properties (e.g. Kang *et al.* 2016). Importantly, the periostracum shows microtopographic differences among mussel species (e.g. Bers *et al.* 2006, 2010; Kang *et al.* 2016). Here, we document the biological diversity of epibiotic bryozoans found on the invasive mussel *M. galloprovincialis*, highlighting the possibility that invasive species, by functioning as basibionts, can serve as a new substratum for epibionts.

Material and methods

Live adults of *Mytilus galloprovincialis* (2958 individuals) were collected between December 2019 and October 2020 from mixed-species mussel patches in the balanoid zone (i.e. the midlittoral zone; Stephenson & Stephenson 1949) at 15 intertidal rocky shore sites during low spring tides across approximately 650 km of the south–southeast coast of South Africa (Fig. 1). The mussels were collected as part of a larger project (Ma *et al.* 2021, 2023) and these included sites toward the eastern limit of distribution of *M. galloprovincialis*, where few mussels were found. From east to west mussels were subsampled/collected from Tenza Beach (32°22'37.8"S, 28°45'06.2"E), Mazeppa Bay (32°28'25.0"S, 28°39'25.5"E), Seagulls (32°38'31.0"S, 28°25'41.5"E), Kei Mouth (32°41'03.4"S, 28°22'59.3"E), Morgan Bay (32°42'41.1"S, 28°20'21.7"E), Kidds Beach (33°08'57.3"S, 27°42'09.2"E), Schoenmakerskop (34°02'27.4"S, 25°31'59.5"E), Jeffreys Bay (34°01'35.5"S, 24°55'52.2"E), St. Francis Bay (34°10'15.7"S, 24°50'06.2"E), and Mossel Bay (34°10'52.4"S, 22°09'29.1"E). No epibiotic bryozoans were found on 371 mussels collected at five sites: Haga Haga (32°45'50.2"S, 28°15'06.8"E), Cintsa West (32°50'42.0"S, 28°07'10.1"E), Glen Gariff (32°53'35.1"S, 28°05'28.9"E), Old Woman's River (33°28'57.5"S, 27°09'04.7"E), and Nature's Valley (33°59'13.9"S, 23°32'51.8"E). To determine the overall prevalence *sensu* Calvo-Ugarteburu & McQuaid (1998), the number of mussels with epibiotic bryozoans was divided by the total number of mussels that were sampled at each site and expressed as a percentage.

A Wild Heerbrugg M8 dissecting microscope was used to examine specimens and separate bryozoan species for taxonomic identification. Specimens were bleached for 15–30 minutes in 20% household bleach and rinsed in distilled water. Subsamples were dried overnight and mounted onto stubs. Imaging was done on uncoated specimens using a Hitachi TM4000Plus scanning electron microscope (SEM). The systematic classification used follows the World Register of Marine Species (WoRMS Editorial Board 2022).

Diagnostic measurements were taken from the SEM images using the software ImageJ (<https://imagej.nih.gov/ij/>). Mean measurements ($\pm$ SD; standard deviation) of up to 20 autozooids (depending on the colony size; N_T is the number of autozooids measured) were tabulated for the following morphological features: autozooid length and width (ZL, ZW) opesia length and width (OpL, OpW); orifice length and width (OL, OW); interzooidal avicularium length and width (AvL, AvW); ovicell length and width (OvL, OvW).

All specimens of *M. galloprovincialis* were dry samples and, along with associated epibiotic bryozoans described and illustrated here, are housed in the Marine Invertebrate collection of the Iziko South African Museum in Cape Town (SAMC), South Africa.

Results

A total of 296 mussels supported epibiotic bryozoans. The overall prevalence of epibiotic bryozoans on mussels was variable among sites, averaging 10% (i.e. 296 of 2958 mussels examined, see Table 1). Table 1 reports from east to west, the sites, number and proportion of mussels with epibiotic bryozoans. Given that detailed taxonomic identification of the epibiotic bryozoans requires the destruction of the mussel shell (substratum), which would interfere with the overall aim of the main project, the subsample size was kept relatively small, with 27 shells examined.

TABLE 1. Presence/absence, given as 1/0 values, of epibiotic bryozoan species on *Mytilus galloprovincialis* mussels collected at sites (from east to west). Abbreviations: N_{epibiont} = number of mussels with epibiotic bryozoans; $N_{\text{basibiont}}$ = total number of mussels sampled; % prevalence = prevalence per site expressed as a percentage.

Area	Sites	Species			$N_{\text{epibiont}}/N_{\text{basibiont}}$	% prevalence
		<i>C. atypica</i> n. sp.	<i>C. hyalina</i>	<i>Hippomonavella</i> sp.		
Eastern Cape	Tenza Beach	1	1	0	1/1	100.00
	Mazeppa Bay	0	1	0	1/2	50.00
	Seagulls	1	0	0	2/2	100.00
	Kei Mouth	1	1	0	3/19	15.79
	Morgan Bay	0	1	0	1/22	4.55
	Haga Haga	0	0	0	0/21	-
	Cintsa West	0	0	0	0/3	-
	Glen Gariff	0	0	0	0/33	-
	Old Woman's River	0	0	0	0/80	-
	Kidds Beach	1	1	0	4/169	2.37
	Schoenmakerskop	1	1	0	14/513	2.73
	Jeffreys Bay	1	0	1	100/241	41.49
	St. Francis Bay	1	1	0	147/720	20.42
Western Cape	Nature's Valley	0	0	0	0/234	-
	Mossel Bay	0	1	0	23/898	2.56
Overall				296/2958	10.01	
				% prevalence		

Eighty-nine bryozoan colonies were observed on the selected 27 shells of *Mytilus galloprovincialis*. Three encrusting cheilostome taxa were identified, namely *Chaperia atypica* n. sp., *Celleporella hyalina* (Linnaeus, 1767) and *Hippomonavella* sp. Additional epibiotic fauna found on some of the *M. galloprovincialis* specimens included barnacles and polychaete tubes.

Chaperia atypica n. sp. and *C. hyalina* were found at most sites in the east (sites from Tenza Beach to St. Francis Bay), but *Chaperia atypica* n. sp. was not found farther west (Mossel Bay; Fig. 1) (Table 1). *Hippomonavella* sp. was found only in Jeffreys Bay in the Eastern Cape (this study, see Table 1) but was previously observed c. 600 km farther west in False Bay (Boonzaaier 2017). The epibiotic bryozoans with the highest abundance was *Chaperia atypica* n. sp. (60 colonies), followed by *C. hyalina* (24 colonies) (Table 2). Only five colonies of *Hippomonavella* sp. were found on just two mussels.

TABLE 2. Total abundance of epibiotic bryozoans in the subset of 27 *Mytilus galloprovincialis* shells studied, and bryozoan colony sizes (in mm) for each species. Abbreviations: TA = total abundance (i.e. number of bryozoan colonies); N_T = number of mussels with epibiotic bryozoans; SD = standard deviation.

Species	TA	N_T	Colony length		Colony width	
			Range	Mean $\pm$ SD	Range	Mean $\pm$ SD
<i>C. atypica</i> n. sp.	60	23	1–30	6 $\pm$ 5	1–19	8 $\pm$ 5
<i>C. hyalina</i>	24	9	1–8	3 $\pm$ 2	1–11	4 $\pm$ 3
<i>Hippomonavella</i> sp.	5	2	1–4	11 $\pm$ 7	6–23	15 $\pm$ 8

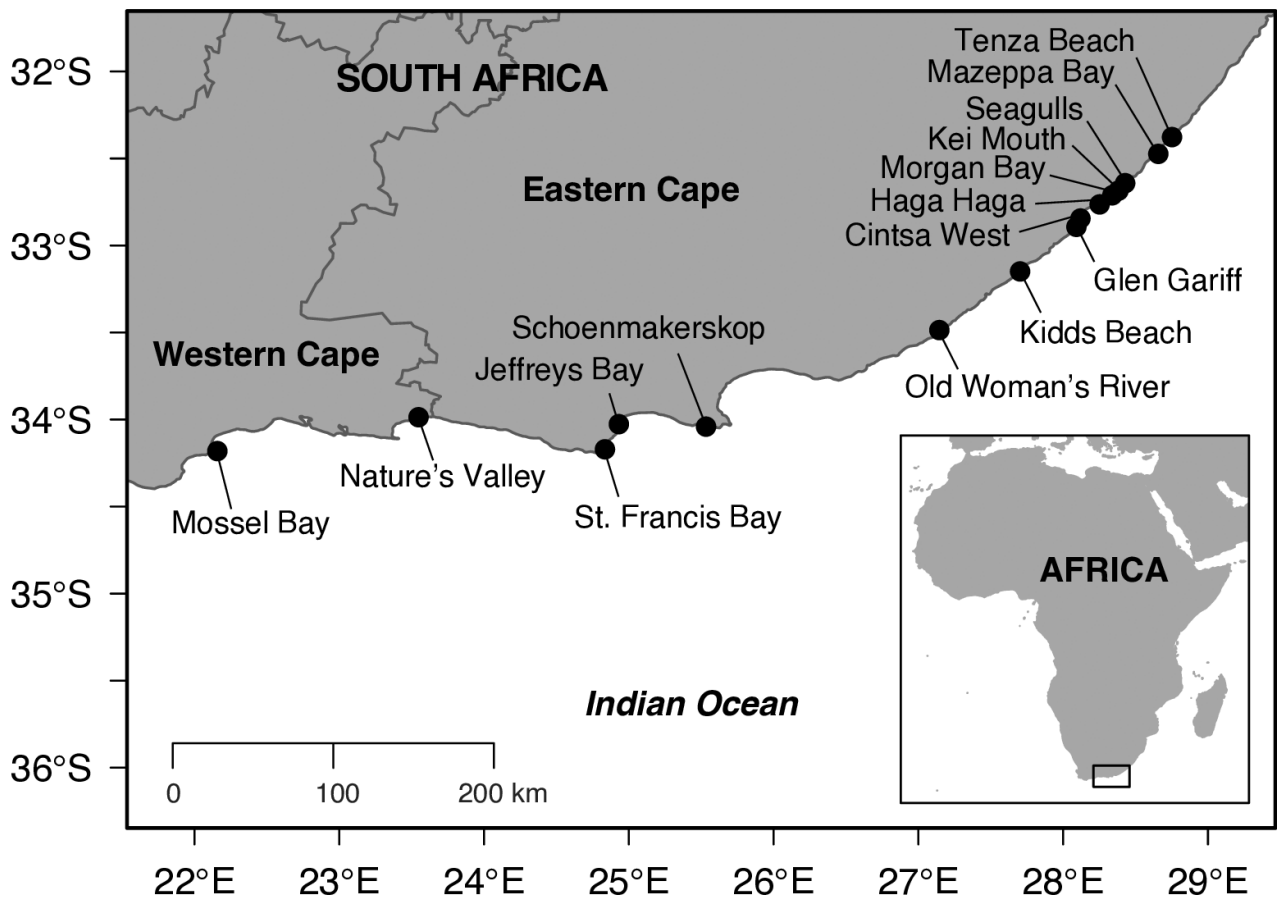


FIGURE 1. Map showing the location of 15 intertidal rocky-shore sites along the south–southeast coast of South Africa where epibiotic bryozoans were sampled on live shells of the mussel *Mytilus galloprovincialis*.

Systematic Account

Class Gymnolaemata Allman, 1856

Order Cheilostomatida Busk, 1852

Suborder Flustrina Smitt, 1868

Superfamily Calloporoidea Norman, 1903

Family Chaperiidae Jullien, 1888

Genus *Chaperia* Jullien, 1881

***Chaperia atypica* n. sp.**

(Fig. 2A–F, Table 3)

<http://zoobank.org/urn:lsid:zoobank.org:act:1D5772C8-99E4-40B7-ADA6-DF404C2F1D0D>

Material examined. *Holotype*: SAMC-A094514 (dry), Seagulls (32°38'31.0"S, 28°25'41.5"E), 8 March 2020. *Paratype*: SAMC-A094513, same details as the holotype. *Other non-type specimens*: SAMC-A094511, Tenza Beach (32°22'37.8"S, 28°45'06.2"E), 29 January 2020. SAMC-A094508 (on the same substratum as *C. hyalina*), SAMC-A094509 (on the same substratum as *C. hyalina*), SAMC-A094512, Kei Mouth (32°41'03.4"S, 28°22'59.3"E), 30

January 2020. SAMC-A094515-18, Kidds Beach (33°08'57.3"S 27°42'09.2"E), 7 March 2020. SAMC-A094520-25, Schoenmakerskop (34°02'27.4"S, 25°31'59.5"E), 5 August 2020. SAMC-A094528, St. Francis Bay (34°10'15.7"S, 24°50'06.2"E), 4 August 2020. SAMC-A094630-34, Jeffreys Bay (34°01'35.5"S, 24°55'52.2"E), 3 August 2020. All material was collected by K.C.K. Ma and is kept dry.

Additional comparative material examined: *Chaperia capensis* (Busk, 1884), SAMC-A026485, station SM 179 (33°30.3'S, 27°22.1'E), off Great Fish River Mouth, Southeast Coast, South Africa, dredged, depth 80 m, 29 May 1978. *Chaperia septispina* Florence, Hayward & Gibbons, 2007, SAMC-A028571, Homestead Plateau, Oudekraal (33°58'90"S, 17°15'00"E), depth 12 m, collected by W. Florence, 30 April 1999.

Etymology. From the Latin *a-* plus *typica*, meaning atypical, not usual, and referring to the atypical diagnostic features noted in this new *Chaperia* compared to other species of the genus.

Diagnosis. Colony encrusting. Cryptocyst typically granular, extensive; gymnocyst reduced or absent. Suborificial occlusor laminae well-developed. Spines on distal rim. Interzooidal avicularia and vestigial ooecia present. Pore chamber windows present.

Description. Colony encrusting, unilaminar, multiserial, usually forming small patches, some colonies up to 30 mm in diameter (see Table 2). Dried (unbleached) material light brown to creamy white. Autozooids arranged quincuncially (Fig. 2A), rounded distally, widest at one third to half zooidal length, with a pair of distal and one lateral pore chamber windows, approximately at autozoid mid-length (Fig. 2B). Cryptocyst well-developed, occupying nearly the total length of autozoid, a raised rim outlining the autozooids, sometimes with the distal part of the opesia raised; frontal surface granular, grain size 4–12 µm in diameter ($N_T = 60$). Reduced or negligible gymnocyst. Opesia eye-shaped with a pair of well-developed occlusor laminae (Fig. 2C); 6–8 oral spines, more commonly seven ($N_T = 20$; length range = 0.17–0.35 mm; 12–28 µm in diameter) distributed evenly around the distal part of the opesia in a shallow arc (Fig. 2E), ending more or less at level with the proximal border of the opesia. Interzooidal avicularia present, twinned or single, each situated disto-medially, projecting into the next distal zooid, distally directed, oval-shaped, the rostrum rounded, with condyles as mandibular pivots (Fig. 2D), more or less situated mid-length of the avicularium. Intramural budding observed (Fig. 2D). Ooecia as a transversely narrow rim on the distal border of opesia with ooecial pore located medio-distally (Fig. 2E, F). Ancestrula not seen.

Remarks. This species is confidently placed in the genus *Chaperia* due to the presence of the eye-shaped opesia not exceeding 50% of the zooidal length, extensive cryptocyst, stout spines around the distal oral border and well-developed occlusor laminae as described in Gordon (1982). Avicularia and ovicells were not reported in the original genus description by Jullien (1881) and subsequent observations made by Gordon (1982). However, interzooidal avicularia and vestigial ooecia were recently observed in some *Chaperia* species from New Zealand (D.P. Gordon, personal communication, 27 July 2022).

These features differ from those of *Chaperiopsis* species that have an oval opesia generally occupying 75% or more of the zooidal length, narrow cryptocyst, negligible occlusor laminae and the presence of frequent small avicularia and prominent hyperstomial ovicells (Gordon 1982).

Chaperia acanthina Lamouroux, 1824 (*C. australis* Jullien, 1881 by original designation, p. 62), originally described from Falkland Islands in the South Atlantic, was also reported from South Africa (O'Donoghue & de Wateville 1937; O'Donoghue 1957; Hayward 1980). *Chaperia atypica* n. sp. differs from this species in the shape of the opesia, in the position and number of spines, and in having a reduced gymnocyst. However, records of *C. acanthina* in South Africa are in need of revision (D.P. Gordon, personal communication, 13 September 2022) and also discussed in Hayward & Cook (1983, p. 19).

Two species of *Chaperia* are known from South Africa, namely *C. capensis* and *C. septispina*. An additional species, *C. polygonia* Kluge, 1914 is known from Antarctica in the Southern Ocean. *Chaperia atypica* n. sp. has 6–8 distolateral spines, while *C. acanthina* has 4–5 distal spines, *C. polygonia* 5–6, *C. septispina* 5–7 and *C. capensis* only two.

This new species is the first *Chaperia* with avicularia and the first South African species with ooecia. Four out of nine *C. atypica* n. sp. colonies examined using SEM (SAMC-A094513, SAMC-A094516, SAMC-A094524 and SAMC-A094525), indicated the presence of avicularia, usually one or two interzooidal avicularia scattered sporadically in a colony of at least 20 autozooids. One colony (SAMC-A094513), with at least 40 autozooids, had four autozooids associated with a single interzooidal avicularium, and one associated with twinned avicularia (see Fig. 2D). These avicularia differ in shape and frequency from those in *Chaperiopsis*: they are rare and with rounded rostra (Fig. 2D), as opposed to the frequent avicularia with triangular rostra and pedunculate avicularia usually

present for example in *Chaperiopsis multifida* (Busk, 1884), a species reported from South Africa (e.g. Marcus 1922; O'Donoghue & de Watterville 1935; Hayward & Cook 1983; Florence *et al.* 2007) and also New Zealand (Gordon 1984). Vestigial ooecia (as seen in Fig. 2E, F) have also been reported from an Australian *Chaperia* sp. (see Cook *et al.* 2018, p. 107, fig. 3.44A) and are likely to be typical of the genus but previously overlooked.

To date, this species has only been observed forming encrusting patches on the invasive mussel *M. galloprovincialis*, distributed on rocky shores from Tenza Beach to St. Francis Bay on the southeast coast of South Africa.

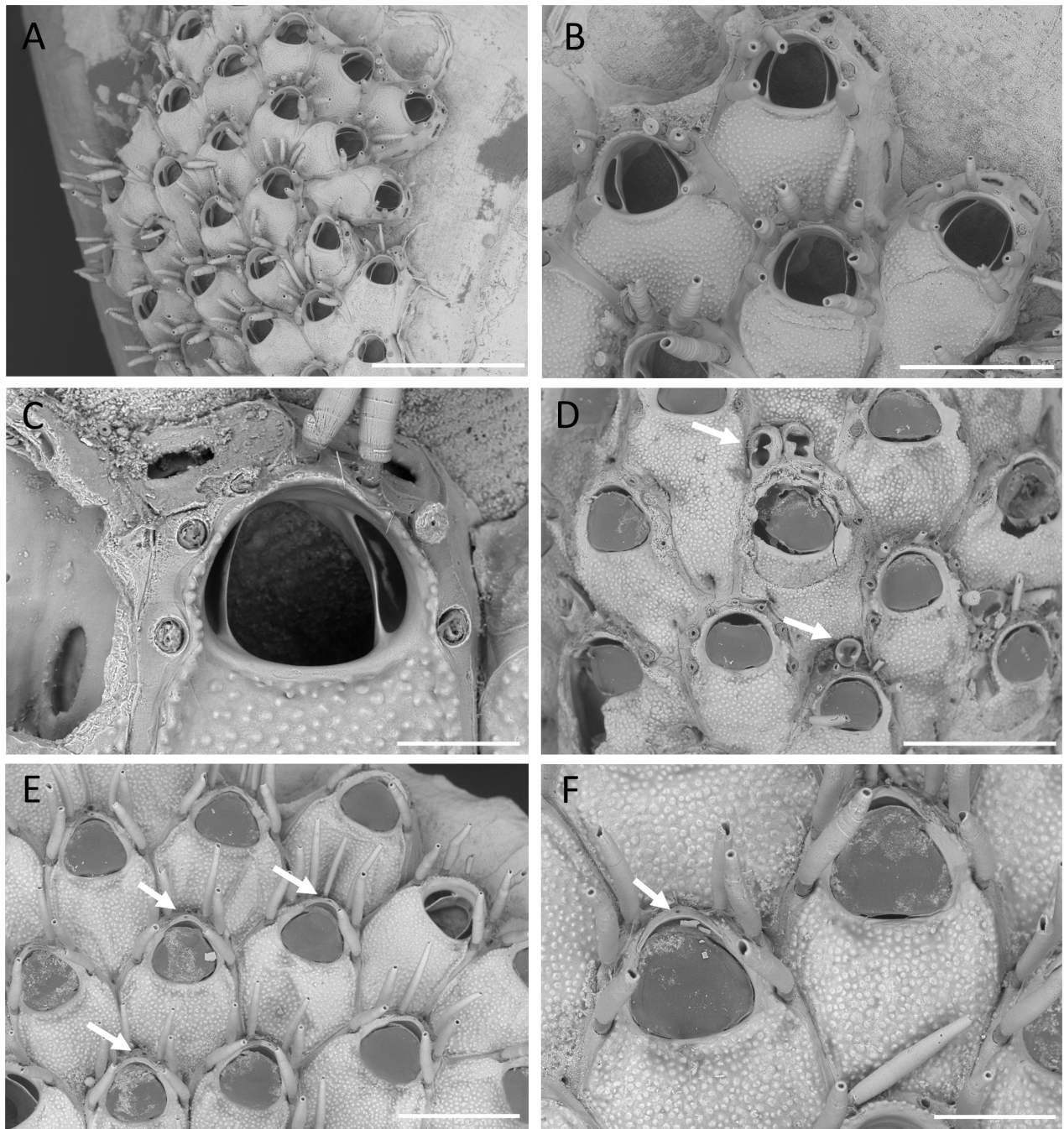


FIGURE 2. *Chaperia atypica* n. sp. A–C. SAMC-A094525. A. General view of the colony. B. Group of zooids, showing pore-chamber windows and spines. C. Close-up of the orifice. D. Paratype, SAMC-A094513. Group of zooids with arrows indicating twinned and single interzooidal avicularia. E, F. Holotype, SAMC-A094514. E. Group of ovicelled zooids with arrows indicating vestigial ooecia. F. Close-up of the vestigial ooecia with an arrow indicating the ooecial pore. Scale bars: A = 1 mm; B = 0.4 mm; C = 0.1 mm; D, E = 0.4 mm; F = 0.2 mm.

TABLE 3. Measurements (in mm) of the holotype SAMC-A094514 of *Chaperia atypica* **n. sp.** and colony SAMC-A094513 for interzooidal avicularia. Abbreviations: N_T = number of structures measured; SD = standard deviation.

Characters	N_T	Range	Mean $\pm$ SD
ZL	20	0.41–0.57	0.51 $\pm$ 0.04
ZW	20	0.26–0.42	0.35 $\pm$ 0.04
OpL	18	0.14–0.19	0.17 $\pm$ 0.01
OpW	15	0.16–0.21	0.19 $\pm$ 0.01
AvL	3	0.08–0.13	0.10 $\pm$ 0.03
AvW	4	0.07–0.11	0.09 $\pm$ 0.02

Superfamily Hippothooidea Busk, 1859

Family Hippothoidae Busk, 1859

Genus *Celleporella* Gray, 1848

Celleporella hyalina (Linnaeus, 1767)

(Fig. 3A–C, Table 4)

Cellepora hyalina Linnaeus, 1767: 1286.

Celleporella hyalina: Gray 1848: 128; Hayward & Ryland 1999: 94, fig. 21; Florence *et al.* 2007: 29, figs. 10D–F.

Schizoporella hyalina: Hincks 1880: 271, pl.18, fig. 1.

Hippothoa hyalina: O'Donoghue 1924: 41; O'Donoghue & de Watteville 1937: 15; O'Donoghue & de Watteville 1944: 424; O'Donoghue 1957: 84; Ryland & Gordon 1977: 38.

Material examined. SAMC-A094507, Morgan Bay (32°42'41.1"S, 28°20'21.7"E), 10 December 2019. SAMC-A094508-09, Kei Mouth (32°41'03.4"S, 28°22'59.3"E), 10–11 December 2019. SAMC-A094510, Mazeppa Bay (32°28'25.0"S, 28°39'25.5"E), 28 January 2020. SAMC-A094518 (on the same substratum as *C. atypica* **n. sp.**) (dry), Kidds Beach (33°08'57.3"S, 27°42'09.2"E), 7 March 2020. SAMC-A094526-27, SAMC-A094528 (on the same substratum as *C. atypica* **n. sp.**), St. Francis Bay (34°10'15.7"S, 24°50'06.2"E), 4 August 2020. SAMC-A094519, Mossel Bay (34°10'52.4"S, 22°09'29.1"E), 26 March 2020. All material was collected by K.C.K. Ma and is kept dry.

Additional comparative material examined: *Celleporella hyalina*, SAMC-A073500 (preserved in ethanol), False Bay, rock dredge, collected by J. Gilchrist, 31 January 1959.

Remarks. *Celleporella hyalina* has been reported to be widely distributed throughout the northern and southern hemispheres (e.g. Hincks 1880; Morris 1980; Moyano 1986). In South Africa, *C. hyalina* has been reported from Saldanha Bay on the west coast to Durban on the east coast (O'Donoghue 1924, 1957; O'Donoghue & de Watteville 1937, 1944; Florence *et al.* 2007).

This species can typically be identified by its elongated, transversely striated frontal wall, often with a lunate umbo below the orifice, orifice of the female zooid wide and short, bearing a porous globular ovicell (Florence *et al.* 2007). The orifice shape of non-ovicelled zooids shown by Florence *et al.* (2007, p. 28, fig. 10E, F) differs from the present material in having a deep U-shaped sinus as opposed to a broad shallow sinus (as seen in Fig. 3B). These differences may reflect inter-lineage or interspecific variation. Morris (1980) also described variation in orifice shape for *C. hyalina* from the Pacific Coast of North America.

Previous reports of *C. hyalina* from South Africa lack morphological measurements useful to quantify intraspecific variation of South African populations. Therefore, re-examining specimens from previous collections would be necessary for this purpose. In the present material, ovicelled zooids ($N_T = 5$; mean length 0.28 mm by mean width 0.18 mm, Table 4) are smaller than non-ovicelled zooids ($N_T = 20$; mean length 0.42 mm by mean width 0.22 mm, Table 4). The ovicells are prominent, globose and irregular in shape, with 5–11 conspicuous pores ($N_T = 10$; mean number of pores = 8 ± 1) distributed over the entire surface as seen in colony SAMC-A094526.

Some studies have shown that *C. hyalina* can be considered as a species complex comprising significant morphological variation (e.g. Morris 1980; Navarrete *et al.* 2005; Grischenko *et al.* 2007). Integrated morphological, molecular, and reproductive-compatibility studies are required to better understand the *C. hyalina* species complex (Dick *et al.* 2005) and to establish species- and population-level diversity within this species complex in South Africa.

Celleporella hyalina forms encrusting patches on mussels (this study), the fronds of kelp, and sometimes pelagic plastic debris in shallow waters (< 30 m) from Saldanha Bay on the west coast to Durban on the east coast (Florence *et al.* 2007).

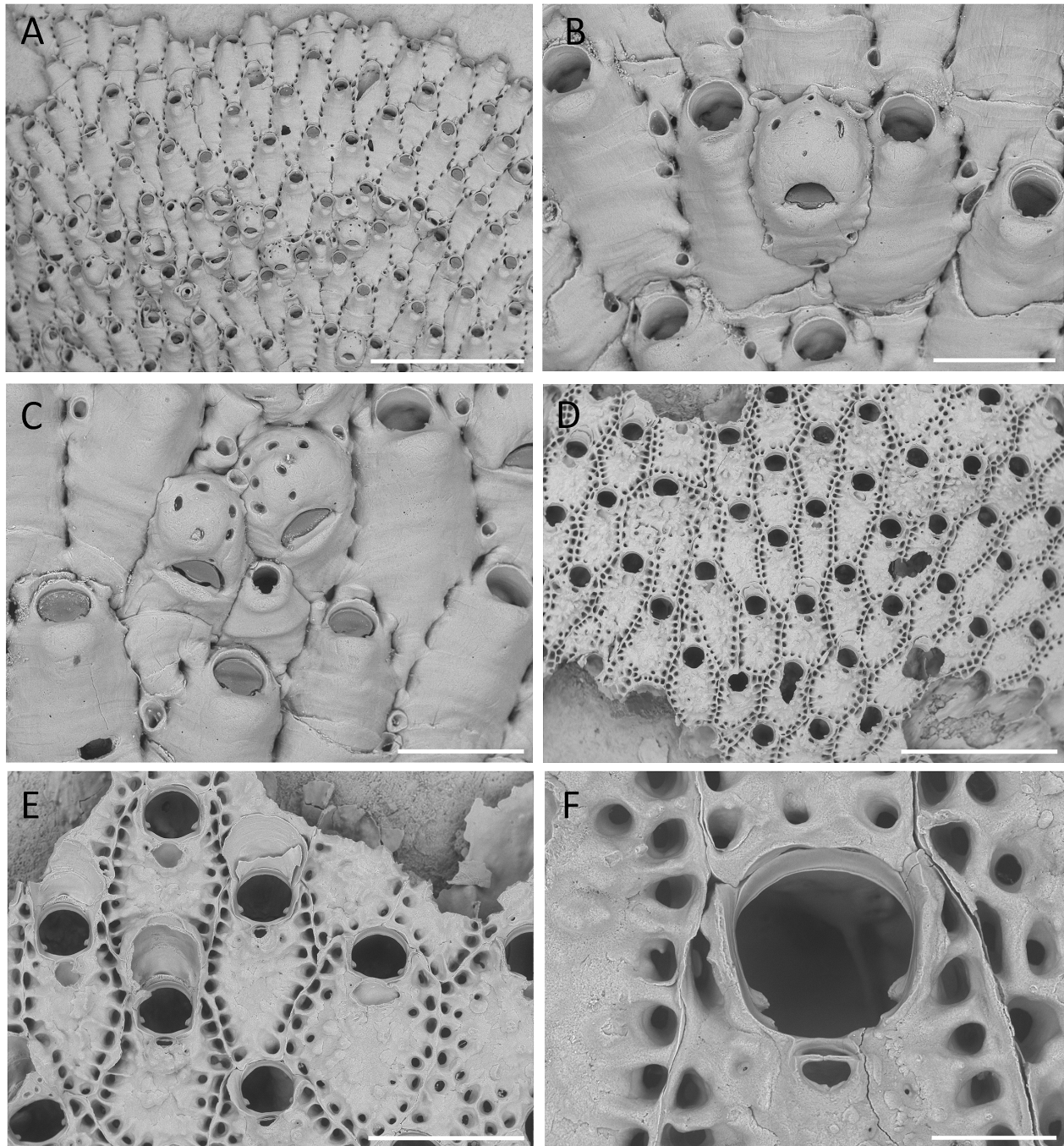


FIGURE 3. A–C. *Celleporella hyalina* SAMC-A094510. A. General view of the colony. B. Single ovicelled zooid. C. Group of ovicelled zooids and a male dwarf zooid. D–F. *Hippomonavella* sp. SAMC-A094529. D. Group of zooids. E. Group of zooids, some with developing ovicells. F. Close-up of the orifice, showing condyles and suboral avicularium. Scale bars: A = 1 mm; B, C = 0.2 mm; D = 1 mm; E = 0.4 mm; F = 0.1 mm.

TABLE 4. Measurements (in mm) of *Celleporella hyalina*. (Z) indicates measurements of non-ovicelled zooids, while (Zo) indicates measurements of zooids with ovicells. All measurements were taken from colony SAMC-A094510 except for measurements of the ovicells, and orifices of non-ovicelled (z) and ovicelled zooids (zo) which were taken from colony SAMC-A094526. Abbreviations: N_T = number of structures measured; SD = standard deviation.

Characters	N_T	Range	Mean $\pm$ SD
ZL	20	0.27–0.57	0.42 $\pm$ 0.09
ZW	20	0.12–0.26	0.22 $\pm$ 0.03
ZoL	5	0.27–0.30	0.28 $\pm$ 0.01
ZoW	5	0.15–0.20	0.18 $\pm$ 0.02
OLz	8	0.07–0.09	0.08 $\pm$ 0.01
OWz	8	0.08–0.09	0.08 $\pm$ 0.00
OLzo	10	0.03–0.04	0.04 $\pm$ 0.00
OWzo	10	0.07–0.09	0.08 $\pm$ 0.01
OvL	10	0.14–0.17	0.16 $\pm$ 0.01
OvW	10	0.18–0.24	0.21 $\pm$ 0.02

Superfamily Smittinoidea Levinsen, 1909

Family Bitectiporidae MacGillivray, 1895

Genus *Hippomonavella* Canu & Bassler in Bassler, 1934

Hippomonavella sp.

(Fig. 3D–F, Table 5)

Material examined. SAMC-A094529, SAMC-A094632, Jeffreys Bay (34°01'35.5"S, 24°55'52.2"E), 3 August 2020. All material was collected by K.C.K. Ma and is kept dry.

Additional comparative material examined: SAMC-A029050, holotype of *Hippomonavella lingulata* Boonzaaier-Davids, Florence & Gibbons, 2020 (preserved in ethanol), station AFR273 A31624 (33°48'24"S, 25°56'48"E), Algoa Bay, Southeast Coast, South Africa, South Coast Demersal Survey, trawl, depth 45 m, 21 April 2011.

Remarks. This species is confidently placed in *Hippomonavella* based on the presence of a typical bitectiporid orifice, median suboral avicularium directed proximally, marginal areolae, and the absence of a lyrula. It differs from known *Hippomonavella* species recorded from South Africa, namely *H. formosa* (MacGillivray, 1887) and *H. lingulata*, in the shape, size, and position of the median suboral avicularium, as well as the orifice shape, suggesting a new record of this genus from South Africa. Comparing measurements of the autozooids showed that *Hippomonavella* sp. (N_T = 20; mean length 0.57 mm by mean width 0.35 mm, Table 5) is smaller than *H. lingulata* (N_T = 14–17; mean length 0.78 mm by mean width 0.50 mm, Boonzaaier-Davids *et al.* 2020, p. 122).

TABLE 5. Measurements (in mm) of *Hippomonavella* sp. from colonies SAMC-A094529 and SAMC-A094632, the latter colony for ovicells. Abbreviations: N_T = number of structures measured; SD = standard deviation.

Characters	N_T	Range	Mean $\pm$ SD
ZL	20	0.47–0.79	0.57 $\pm$ 0.09
ZW	20	0.25–0.53	0.35 $\pm$ 0.06
OL	20	0.10–0.14	0.11 $\pm$ 0.01
OW	20	0.12–0.15	0.13 $\pm$ 0.01
OvL	13	0.18–0.21	0.19 $\pm$ 0.01
OvW	13	0.19–0.24	0.21 $\pm$ 0.01

In this study, five colonies of *Hippomonavella* **sp.** were found on two individuals of *M. galloprovincialis*; it has previously been found forming encrustations on other shells and rocks in c. 600 km farther west False Bay (Boonzaaier 2017).

A separate manuscript revising the superfamily Smittinoidea in South Africa is in preparation and will include the description of this new species, the type material of which will be selected from samples from False Bay sites that yielded better preserved specimens.

Discussion

The role of basibionts and epibionts in ecosystem functioning (e.g. habitat formation and modification) has been well documented and includes systems ranging from estuaries to subtidal rocky reefs, as well as from tropical to polar regions (Thomsen *et al.* 2010). In the present study, we report on epibiotic bryozoans on the invasive populations of *Mytilus galloprovincialis* from the intertidal rocky community of South Africa and document three encrusting cheilostome bryozoan taxa, including a new species. We report and describe the first known aviculiferous *Chaperia* species in South Africa, *Chaperia atypica* **n. sp.**, with vestigial ooecia. Avicularia and vestigial ooecia have been previously observed in closely related *Chaperia* species from New Zealand (D.P. Gordon, personal communication, 27 July 2022).

While a separate manuscript is being prepared to describe the new species of *Hippomonavella* based on specimens previously collected from False Bay (Boonzaaier 2017), it is notable that its range extends approximately 850 km east, from False Bay (Western Cape) to Jeffreys Bay (Eastern Cape).

As part of another study investigating the extent and distribution range of another invasive mussel *Semimytilus patagonicus* (Hanley, 1843), previously known as *Semimytilus algosus* (Gould, 1850), found in South Africa (Ma *et al.* 2020; Ma *et al.* in press), epibiotic bryozoans were also detected on its shells. Over 1500 specimens were collected between Hondeklip Bay on the west coast and Hermanus on the south coast in August, September, and October 2021 (K.C.K. Ma, personal observation, 25 November 2022). Only two shells of *S. patagonicus* (overall prevalence 0.10%) were encrusted by colonies of *C. hyalina* (K.C.K. Ma, personal observation, 25 November 2022) as opposed to 10% of *M. galloprovincialis* shells (this study). This might indicate that bryozoan larvae have potentially a substratum preference, favouring *M. galloprovincialis*. This may warrant further investigation of mussel shell chemistry and features, for example surface texture (e.g. smooth, striated, or rough; see Bers *et al.* 2010) or a reflection of differing biogeographic patterns between the west and south coasts of South Africa (e.g. Ndhlovu *et al.* 2019). Moreover, though epibiotic bryozoans have been well-documented on the native brown mussel, *Perna perna* (Linnaeus, 1758) (e.g. de Sá *et al.* 2007; Lins & Rocha 2022), the diversity and prevalence of epibiotic bryozoans associated with it in South Africa have not yet been investigated.

In South Africa, at least 282 extant marine species of erect and encrusting bryozoans are known and, among these, six are non-indigenous species (NIS) (Florence *et al.* 2007; Griffiths *et al.* 2010; Boonzaaier-Davids *et al.* 2020). These NIS include *Bugula neritina* (Linnaeus, 1758), *Bugulina flabellata* (Thompson in Gray, 1848), *Conopeum seurati* (Canu, 1928), *Cryptosula pallasiana* (Moll, 1803), *Virididentula dentata* (Lamouroux, 1816), and *Watersipora subtorquata* (d'Orbigny, 1852) (Mead *et al.* 2011; Malherbe & Samways 2014; Robinson *et al.* 2020), but their extent in both area and level of invasiveness in South Africa have not yet been quantified (Boonzaaier 2017). The role in which invasive and native basibionts provide habitat for other invasive and native epibiotic bryozoans is largely unexplored in South Africa.

In addition to exhibiting high diversity (including a high number of endemic species; see Boonzaaier-Davids *et al.* 2020), bryozoans in South Africa are involved in some unique interspecific relationships. These include the symbiosis between the encrusting ctenostome *Alcyonidium nodosum* and its subtidal whelk host (Barkai & McQuaid 1988; Gray *et al.* 2005; Ryland 2001). Some South African species are exclusively epibiotic, such as the cheilostome *Hippothoa musivaria* Hayward & Fordy, 1982, which occurs as an epizooite on the erect cheilostome *Adeonella pluscula* Hayward, 1988, growing between the frontal walls of adjacent autozooids (Hayward & Fordy 1982; Florence *et al.* 2007). In particular, the small zooidal size and inconspicuous runner-like colonies of *H. musivaria* suggest that host-specific bryozoan species could be easily overlooked.

Many mussel species are ecosystem engineers, producing shells that introduce complexity into benthic communities and increase species richness (Gutiérrez *et al.* 2003; Borthagaray & Carranza 2007; Sousa *et al.* 2009). Aggregations of mussels create a more stable habitat through the modification of hydrodynamics and the

physical formation of interstitial spaces within mussel patches, and this can facilitate recruitment of conspecifics and other organisms as well as providing a refuge for mobile species from desiccation and predators (Gutiérrez *et al.* 2003, 2019; Palomo *et al.* 2007; Lathlean & McQuaid 2017). Within South Africa and Namibia, the invasive mussels *M. galloprovincialis* and *S. patagonicus* act as habitat forming species, altering the composition, abundance, and structure of intertidal communities by re-engineering the habitat (Robinson *et al.* 2007; Lathlean *et al.* 2016; Sadchatheeswaran *et al.* 2018; Nicastro *et al.* 2020). In addition to native species (e.g. *P. perna*; Puccinelli & McQuaid 2021), our data suggest that shells of invasive mussels can provide a new substratum for other sessile species (typically barnacles, but also bryozoans and polychaetes) to settle and grow on. In particular, we revealed some of the diversity of epibiotic bryozoans in association with an invasive mussel host by resolving the taxonomic identities of at least three bryozoan taxa – a common but newly described species (*Chaperia atypica* n. sp.), a relatively common species (*Celleporella hyalina*), and an uncommon and as yet undescribed species (*Hippomonavella* sp.).

The sample size was kept small, because detailed taxonomic identifications of the epibiotic bryozoans required destroying the mussel shell (substratum) and would interfere with the objectives of the main project (Ma *et al.* 2021, 2023). Therefore, the diversity of epibiotic bryozoans could be much higher than observed in this study. This paper serves as a good first step in identifying a previously unexplored or neglected area of research within South Africa and require further investigation on epibiont- and basibiont composition and diversity.

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